WEEK 1: Networking Fundamentals | Days: 1-7

Day 3 Tasks—IP Addressing

- **TASK: Check your machine's IP using ipconfig (Windows) / Ifconfig (Linux)?**

IP Config Output

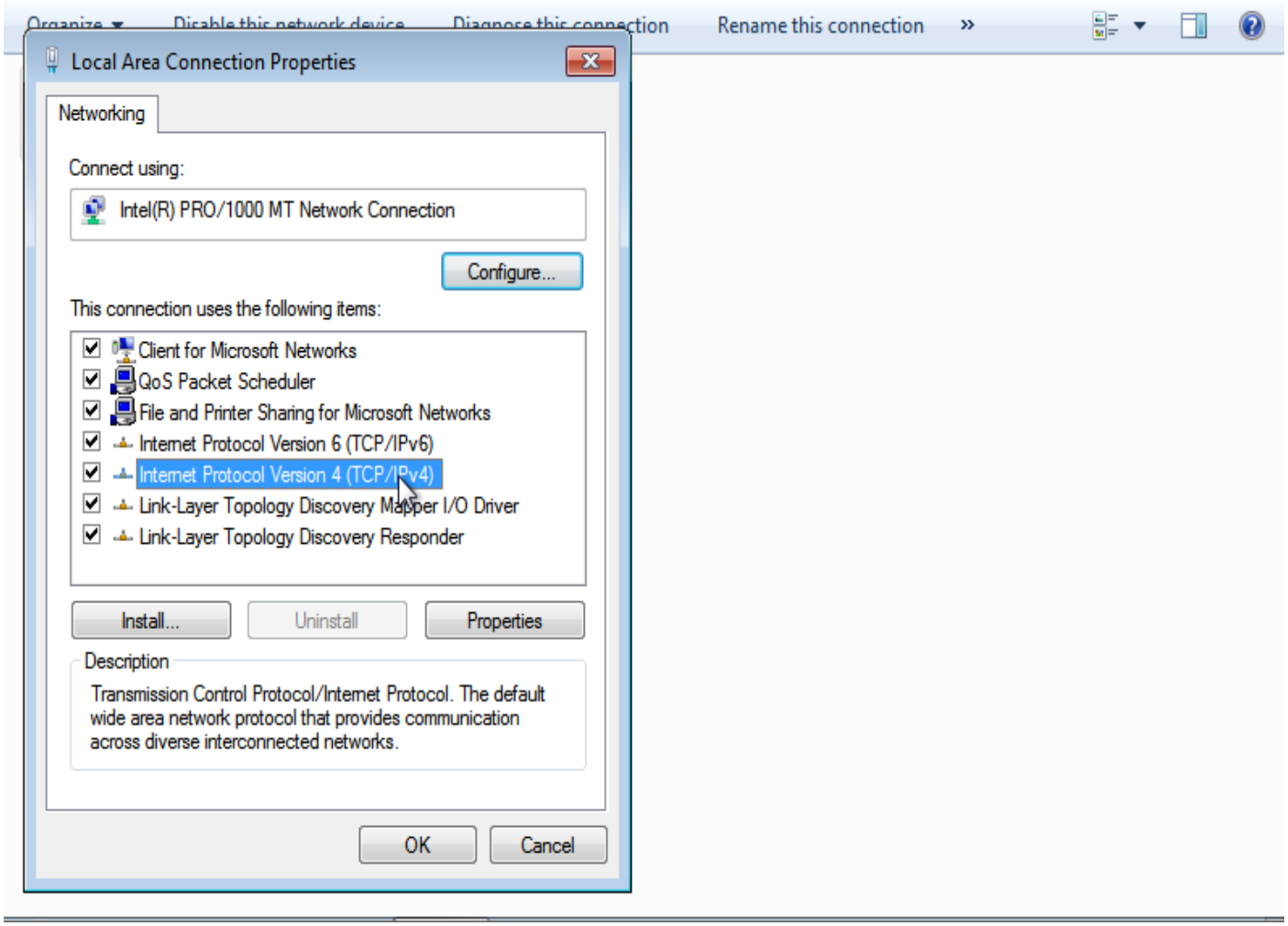
```
C:\Users\SMI>ipconfig

Windows IP Configuration


Ethernet adapter Ethernet:

   Media State . . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Ethernet adapter vEthernet (Default Switch):

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::7818:31c5:5cbc:de43%58
   IPv4 Address. . . . . . . . . . . : 172.25.240.1
   Subnet Mask . . . . . . . . . . . : 255.255.240.0
   Default Gateway . . . . . . . . . :

Ethernet adapter VirtualBox Host-Only Network:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::6f8c:7df0:8963:38ef%8
   IPv4 Address. . . . . . . . . . . : 192.168.56.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :

Wireless LAN adapter Local Area Connection* 1:

   Media State . . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:

   Media State . . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
```

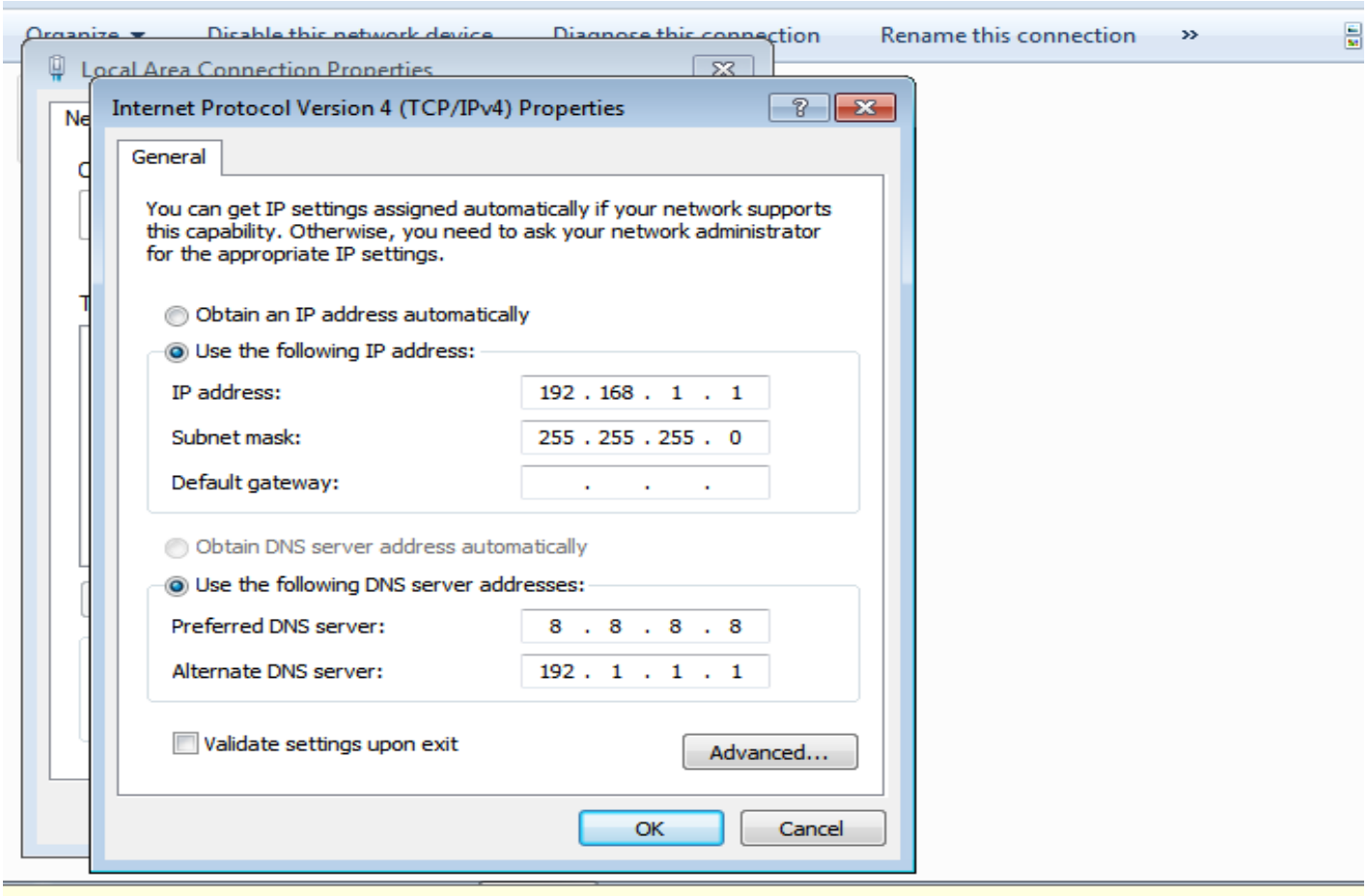
```
Ethernet adapter VMware Network Adapter VMnet1:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::f84b:b209:2ba7:963d%21
   IPv4 Address. . . . . . . . . . . : 192.168.109.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :

Ethernet adapter VMware Network Adapter VMnet8:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::8b0:1c80:db7f:d7d7%6
   IPv4 Address. . . . . . . . . . . : 192.168.117.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::902c:32d7:8933:12fd%13
   IPv4 Address. . . . . . . . . . . : 10.71.253.8
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : 10.71.253.25

Ethernet adapter Bluetooth Network Connection:

   Media State . . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
```

Ifconfig Kali Linux output

```
┌──(kali㉿kali)-[~/Desktop]
└─$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.117.128  netmask 255.255.255.0  broadcast 192.168.117.255
        inet6 fe80::84a9:d2d3:bf5f:5a19  prefixlen 64  scopeid 0x20<link>
        ether 00:0c:29:11:31:ed  txqueuelen 1000  (Ethernet)
        RX packets 31184  bytes 45109100 (43.0 MiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 9479  bytes 579114 (565.5 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 8  bytes 480 (480.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 8  bytes 480 (480.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

- **TASK: Identify whether your IP is private or public and explain why.**





Based on the Windows ipconfig output.

IP address (192.168.1.1) is PRIVATE

Why is it private?

Because it falls within the **Class C private IP range**: `192.168.0.0 – 192.168.255.255`

Additional Observations:

Setting	Value	Analysis
IPv4 Address	192.168.1.1	✅ Private IP
Subnet Mask	255.255.255.0	Standard home/office network
Default Gateway	(blank)	Tis device might be the router itself

Above IP is private because:

1. It starts with 192.168 — reserved for private networks by IANA.
2. It cannot be routed on the public internet.
3. Any device with this IP exists only behind a NAT boundary.

Public IP (what the internet sees) would be whatever IP your ISP assigned to modem/router — completely different from this 192.168.1.1.

➢ TASK: IPv4 address structure (octets, binary, decimal)?

Here is the IPv4 address structure in simple text format.

IPv4 Address Structure (Simple Explanation).

An IPv4 address is a 32-bit number, but we write it as 4 smaller numbers separated by dots.

The 4 Octets

- The address is made of 4 parts called "octets" (because each part has 8 bits).
- Example:192.168.1.1
  - Octet 1: 192
  - Octet 2: 168
  - Octet 3: 1
  - Octet 4: 1

Decimal vs. Binary

- Computers read binary (0s and 1s), but we use decimal (0-9) for simplicity.
- Each octet in decimal ranges from 0 to 255.
- Each octet in binary is 8 digits long (e.g., 192 = `11000000).

Why 0 to 255?

- Because 8 bits can create 256 possible combinations (2^8 = 256).
- Counting starts at 0, so the max number is 255.

Full Binary Example

- Decimal: `192.168.1.1`
- Binary: `11000000.10101000.00000001.00000001`

Summary

- **Structure:** 4 numbers (0-255) separated by dots.
- **Total size:** 32 bits.
- **Purpose:** Every device on a network gets one to send and receive data.

➢ TASK: Address classes: Class A, B, and C ranges.

Here are the IPv4 address classes (A, B, and C) in simple text format.

Class A

Range: 1.0.0.0 to 126.255.255.255

First octet: 1 to 126.

Default subnet mask: 255.0.0.0 (or /8).

Network/Host: First octet = network | Last 3 octets = host.

Use: Very large networks (like big companies or ISPs).

Private range: 10.0.0.0 to 10.255.255.255.

Class B

Range: 128.0.0.0 to 191.255.255.255.

First octet: 128 to 191.

Default subnet mask: 255.255.0.0 (or /16).

Network/Host: First 2 octets = network | Last 2 octets = host.

Use: Medium to large networks (like universities).

Private range: 172.16.0.0 to 172.31.255.255.

Class C

Range: 192.0.0.0 to 223.255.255.255.

First octet: 192 to 223.

Default subnet mask: 255.255.255.0 (or /24).

Network/Host: First 3 octets = network | Last octet = host.

Use: Small networks (like home or small business).

Private range: 192.168.0.0 to 192.168.255.255.

Quick Way to Identify Class

- **Look at the first octet only:**
  - 1 to 126 = Class A
  - 128 to 191 = Class B
  - 192 to 223 = Class C

Special Note

- 127.x.x.x is reserved for loopback (localhost) — not used for regular networks.
- Class D (224-239) is for multicast.
- Class E (240-255) is reserved for experimental use.

(The End)